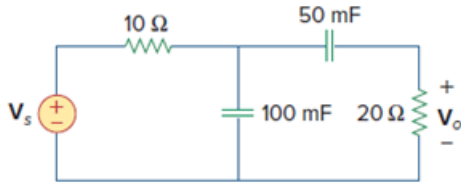


Problem

For the circuit shown in Fig. 14.73, find $\mathbf{H}(s) = \mathbf{V}_o(s)/\mathbf{V}_s(s)$.

Figure 14.73:



Step-by-step solution

Step 1 of 5

Refer to circuit diagram in Figure 14.73 in the text book.

Calculate the value of 100 mF capacitance in frequency domain.

$$X_{100 \text{ mF}} = \frac{1}{j\omega(100 \times 10^{-3})}$$

$$= \frac{10}{j\omega}$$

Calculate the value of 50 mF capacitance in frequency domain.

$$X_{50 \text{ mF}} = \frac{1}{j\omega(50 \times 10^{-3})}$$

$$= \frac{20}{j\omega}$$

Step 2 of 5

Draw the frequency domain circuit.

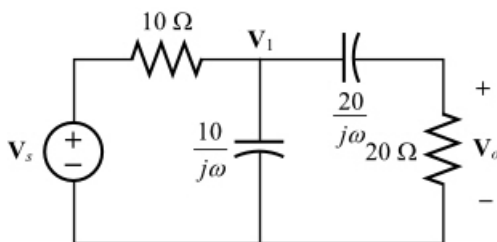


Figure 1

Step 3 of 5

Apply Kirchhoff's current law at node $\mathbf{V}_1$.

$$\begin{aligned}\frac{\mathbf{V}_1 - \mathbf{V}_s}{10} + \frac{\mathbf{V}_1}{\frac{10}{j\omega}} + \frac{\mathbf{V}_1}{\frac{20}{j\omega} + 20} &= 0 \\ \left(\frac{1}{10} + \frac{j\omega}{10} + \frac{j\omega}{20 + j20\omega} \right) \mathbf{V}_1 &= \frac{\mathbf{V}_s}{10} \\ \left(\frac{(1 + j\omega)(20 + j20\omega) + j10\omega}{(10)(20 + j20\omega)} \right) \mathbf{V}_1 &= \frac{\mathbf{V}_s}{10} \\ \left(\frac{20 + j20\omega + j20\omega - 20\omega^2 + j10\omega}{(20 + j20\omega)} \right) \mathbf{V}_1 &= \mathbf{V}_s \\ \mathbf{V}_1 &= \left(\frac{20 + j20\omega}{20 + j50\omega - 20\omega^2} \right) \mathbf{V}_s\end{aligned}$$

Step 4 of 5

Write the expression for output voltage using voltage division rule.

$$\begin{aligned}\mathbf{V}_o &= \left(\frac{20}{\frac{20}{j\omega} + 20} \right) \mathbf{V}_1 \\ \text{Substitute } \left(\frac{20 + j20\omega}{20 + j50\omega - 20\omega^2} \right) \mathbf{V}_s &\text{ for } \mathbf{V}_1. \\ \mathbf{V}_o &= \left(\frac{j20\omega}{20 + j20\omega} \right) \left(\frac{20 + j20\omega}{20 + j50\omega - 20\omega^2} \right) \mathbf{V}_s \\ \frac{\mathbf{V}_o}{\mathbf{V}_s} &= \frac{j20\omega}{20 + j50\omega - 20\omega^2} \\ \mathbf{H}(j\omega) &= \frac{j20\omega}{20 + j50\omega - 20\omega^2}\end{aligned}$$

Step 5 of 5

Substitute s for $j\omega$ in the $\mathbf{H}(j\omega)$ equation.

$$\begin{aligned}\mathbf{H}(s) &= \frac{20s}{20 + 50s + 20s^2} \\ &= \frac{s}{s^2 + 2.5s + 1}\end{aligned}$$

Therefore, the transfer function, $\mathbf{H}(s)$ is $\boxed{\frac{s}{s^2 + 2.5s + 1}}$.